A MULTIMODAL MACHINE LEARNING APPROACH FOR SCRUB TYPHUS CLASSIFICATION USING CLINICAL AND SKIN IMAGE DATA

A project report submitted in partial fulfillment of the requirements for the award of the Degree of

MASTER OF COMPUTER APPLICATIONS

Submitted by

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This is to certify that the project work entitled **"A MULTIMODAL MACHINE LEARNING APPROACH FOR SCRUB TYPHUS CLASSIFICATION USING CLINICAL AND SKIN IMAGE DATA"** has been submitted by **P. Renuka Chowdary (24M11MC134)** in partial fulfilment of the requirements for the award of the degree of Master of Computer Applications.

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DECLARATION

I hereby declare that the project entitled **"A MULTIMODAL MACHINE LEARNING APPROACH FOR SCRUB TYPHUS CLASSIFICATION USING CLINICAL AND SKIN IMAGE DATA"** is original and has been submitted to ADITYA UNIVERSITY, Surampalem, in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications. I further declare that this project work has not been submitted in full or in part for the award of similar or any other degree in this or any other institution and confirm that this work complies with the institutional Academic Integrity and AI Usage Policy.

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ABSTRACT

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Scrub typhus is a vector-borne infectious disease caused by Orientia tsutsugamushi and transmitted through infected chigger mite bites, and it is widely prevalent in tropical and subtropical regions, especially in Asia. The disease commonly presents symptoms such as fever, headache, myalgia, eschar, chills, sweating, and vomiting, and if not diagnosed early it can lead to serious complications including multi-organ failure and increased mortality. Early detection is challenging because these symptoms are similar to other febrile illnesses such as dengue, malaria, and typhoid, making accurate diagnosis difficult. Traditional diagnostic approaches mainly rely on clinical observation and laboratory tests, which may sometimes be time-consuming and prone to misdiagnosis. To address these challenges, this study proposes a multimodal machine learning approach for scrub typhus classification using clinical symptom data and skin image data. The clinical dataset includes symptoms such as fever, headache, myalgia, eschar, chills, sweating, and vomiting, which are analyzed using machine learning algorithms including Support Vector Classifier (SVC) and Random Forest to classify the presence of the disease. In addition, deep learning techniques are applied to analyze skin lesion images for identifying eschar patterns associated with scrub typhus. To enhance model interpretability, SHAP (SHapley Additive Explanations) is used to evaluate the contribution of each clinical feature in the prediction process, while Grad-CAM++ visualization is used to highlight important regions in skin images influencing the model's decision. The predictions from both clinical and image-based models are integrated to improve the overall classification performance. However, the proposed methodology has certain limitations such as limited dataset size, possible data imbalance, and dependency on image quality. Despite these challenges, the multimodal approach can support early detection and assist healthcare professionals in accurate and timely diagnosis of scrub typhus.

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CHAPTER 1

INTRODUCTION

1.INTRODUCTION

1.1 Information to Scrub Typhus Identification

Scrub typhus is a vector-borne infectious disease caused by the bacterium Orientia tsutsugamushi and transmitted through infected chigger mite bites. It is commonly found in tropical and subtropical regions, especially in Asia, and poses a serious public health concern. The disease presents symptoms such as fever, headache, myalgia, eschar, chills, sweating, and vomiting, which are similar to other febrile illnesses, making early diagnosis difficult. To address this challenge, this project proposes a multimodal machine learning approach for scrub typhus classification using both clinical symptom data and skin image data. Machine learning algorithms such as Support Vector Classifier (SVC) and Random Forest are used to analyze clinical features, while deep learning techniques are applied to detect eschar patterns from skin images. In addition, explainable AI techniques such as SHAP are used to identify the importance of clinical features, and Grad-CAM++ is used to visualize important regions in skin images influencing the model's prediction. This approach helps improve the accuracy and reliability of scrub typhus detection and supports healthcare professionals in early diagnosis and treatment.

1.2 Motivation

Scrub typhus is a serious infectious disease that requires early diagnosis to prevent severe complications. However, its symptoms are similar to other febrile diseases, making accurate detection difficult. Traditional diagnostic methods can be time-consuming and sometimes inaccurate. Therefore, this project is motivated by the need to develop a multimodal machine learning system that uses clinical symptoms and skin image data for better classification of scrub typhus. The use of machine learning models and explainable AI techniques such as SHAP and Grad-CAM++ helps improve prediction accuracy and assists healthcare professionals in early diagnosis.

1.3 An Overview of Scrub Typhus Classification

Scrub typhus, caused by Orientia tsutsugamushi and transmitted by larval chigger mites, is a major public health concern across the Asia-Pacific "tsutsugamushi triangle." It is widely endemic in countries like India, China, Japan, Korea, and Southeast Asia. In India, cases are widespread, with significant reporting from Tamil Nadu, Karnataka, Kerala, Andhra Pradesh, Assam, Meghalaya, Nagaland, and Odisha, particularly during and after the monsoon season, as reported in studies referenced by the World Health Organization (WHO).

Epidemiological Distribution of Scrub Typhus

Region	**Gender Most Affected**	**Season / Time Period**	**Percentage / Prevalence**
India	Mostly males (due to agricultural work exposure)	Post-monsoon (July – November)	~40% prevalence
Japan	Both genders affected	Autumn season	~68.4% prevalence
Sri Lanka	Both genders affected	Monsoon season	~38.7% prevalence
South Korea	Slightly higher in males	Autumn season	~35.2% prevalence
North-East India	Rural male population more affected	Cooler months	Increasing number of reported cases

Table 1.3 Epidemiological Distribution

Common Clinical Presentation:

The illness presents as an acute febrile infection with symptoms including fever, headache, myalgia (muscle pain), and eschar (a black, scab-like lesion at the site of the mite bite). The disease often presents as an undifferentiated febrile illness, making diagnosis challenging without high clinical suspicion. The disease is considered a significant, often neglected cause of acute febrile illness in the region.

1.4 Objective

The objective of this project is to develop a multimodal machine learning model for scrub typhus classification using clinical symptoms and skin image data. It uses algorithms such as Support Vector Classifier (SVC) and Random Forest for clinical analysis, along with deep learning for skin image detection. Explainable AI techniques like SHAP and Grad-CAM++ are used to interpret the model and improve prediction reliability for early diagnosis.

1.5 Organization of the Project

The project report is organized into several chapters to provide a structured explanation of the research and implementation process.

Chapter 1: Introduction – This chapter gives a basic overview of scrub typhus, the motivation, and the objectives of the project.

Chapter 2: Literature Survey – This chapter discusses previous research and existing methods related to scrub typhus detection and disease classification.

Chapter 3: Methodology – This chapter explains the approach used in the project, including data collection, preprocessing, and the multimodal machine learning method.

Chapter 4: Implementation – This chapter describes how the system was implemented using Python and Google Colab.

Chapter 5: Module Description – This chapter explains the different modules of the system such as clinical data analysis, image classification, and multimodal prediction.

Chapter 6: Results and Discussion – This chapter presents the results obtained from the models and discusses the performance of the system.

Chapter 7: Conclusion and Future Work – This chapter summarizes the project and suggests possible improvements for future work.

CHAPTER 2

LITERATURE SURVEY

2. LITERATURE SURVEY

Scrub typhus is an infectious disease caused by the bacterium Orientia tsutsugamushi and transmitted through infected chigger mites. It is commonly found in the Asia-Pacific region and is considered a major public health concern in many developing countries. Early detection of scrub typhus is important because delayed diagnosis may lead to severe complications such as organ failure, respiratory distress, and even death. However, diagnosing scrub typhus is difficult because the symptoms often resemble other febrile illnesses such as dengue, malaria, and typhoid. Researchers have explored several diagnostic approaches including laboratory tests, molecular detection methods, and computational techniques to improve disease identification.

Kala and Sharma [1] conducted a comprehensive review on the diagnosis of scrub typhus and highlighted the limitations of conventional diagnostic methods. Their study discussed commonly used techniques such as the Weil-Felix test, enzyme-linked immunosorbent assay (ELISA), and polymerase chain reaction (PCR). Although these laboratory methods provide reliable results, they require specialized equipment and trained personnel, which may not always be available in rural healthcare facilities. The authors emphasized the need for improved diagnostic systems that can assist healthcare professionals in detecting the disease more efficiently.

Liao et al. [2] proposed a diagnostic platform based on recombinant antigens for the detection of scrub typhus infection. Their research developed an improved ELISA-based detection method that showed high sensitivity and specificity when compared to traditional serological tests. The study demonstrated that recombinant antigen technology could significantly improve the reliability of scrub typhus diagnosis.

Kim et al. [3] introduced a highly sensitive molecular diagnostic platform for detecting scrub typhus infection. Their system combined nucleic acid extraction and molecular detection techniques to identify the presence of Orientia tsutsugamushi in patient samples. The proposed method provided rapid detection and improved accuracy compared to traditional laboratory techniques. This research highlighted the importance of molecular diagnostic approaches in improving early detection of infectious diseases.

Lu et al. [4] developed a machine learning-based prognostic model to predict the severity of scrub typhus infection. The model analyzed clinical data such as patient symptoms, laboratory results, and demographic information to predict disease progression. The results of the study showed that machine learning algorithms could effectively identify patterns in patient data and assist healthcare professionals in predicting disease severity. This research demonstrated the potential of artificial intelligence in improving medical decision-making.

Kanchanapiboon et al. [5] explored the application of deep learning techniques for analyzing medical images related to scrub typhus infection. Their study used computer vision algorithms to detect visual patterns in dermatological images associated with the disease. Deep learning models were able to automatically extract important features from images and classify them based on disease characteristics. The results showed that deep learning models can support doctors in identifying visual symptoms of scrub typhus.

Li et al. [6] proposed an electrochemical biosensor for detecting scrub typhus DNA using nanomaterial-based sensors. The biosensor system provided a rapid and highly sensitive detection method capable of identifying the presence of the pathogen in biological samples. This research demonstrated that biosensor technologies could provide faster diagnostic solutions compared to traditional laboratory tests.

Recent advancements in artificial intelligence and machine learning have improved disease diagnosis by analyzing large datasets and identifying complex patterns. Deep learning models such as CNN are widely used for medical image analysis. However, many existing methods use only a single type of data. This project proposes a multimodal machine learning approach that combines clinical data and skin lesion images for scrub typhus classification using explainable AI techniques like SHAP and Grad-CAM++.

Proposed Work

Ref No	Author /Year	Method Used	Data Used	Key Contributon
[1]	Kala & Sharma (2020)	Serological Tests	Clinical samples	Reviewed traditional Scrub Typhus diagnostic methods
[2]	Liao et al. (2021)	ELISA Diagnostic Platform	Blood samples	Improved Scrub Typhus detection accuracy using antigen-based tests
[3]	Kim et al. (2024)	Molecular Diagnostic Platform	Clinical samples	Developed rapid pathogen detection using molecular techniques
[4]	Lu et al. (2025)	Machine Learning Model	Clinical data	Predicted severity of Scrub Typhus using clinical features
[5]	Kanchanapiboon et al. (2024)	Deep Learning (CNN)	Medical images	Applied deep learning for medical image analysis
[6]	Li et al. (2021)	Biosensor Technology	DNA samples	Rapid detection of Scrub Typhus using biosensors
	Proposed Work	Multimodal Machine Learning (Random Forest + CNN)	Clinical Data + Skin Images	Develops an automated system for scrub typhus detection using clinical symptoms and skin image analysis

Table 2 Comparison with Proposed work

CHAPTER 3

METHODOLOGY

3. METHODOLOGY

3.1 System Architecture

System architecture describes the structure and workflow of the proposed scrub typhus classification system. It explains how different components such as clinical data, skin image data, preprocessing, and machine learning models work together to produce the final result. In this system, clinical symptoms are analyzed using SVC and Random Forest algorithms, while skin images are analyzed using a CNN model. The outputs from both analyses are combined using a multimodal approach to improve prediction accuracy. Explainable AI techniques such as SHAP and Grad-CAM++ are used to interpret the results and highlight important features influencing the prediction.

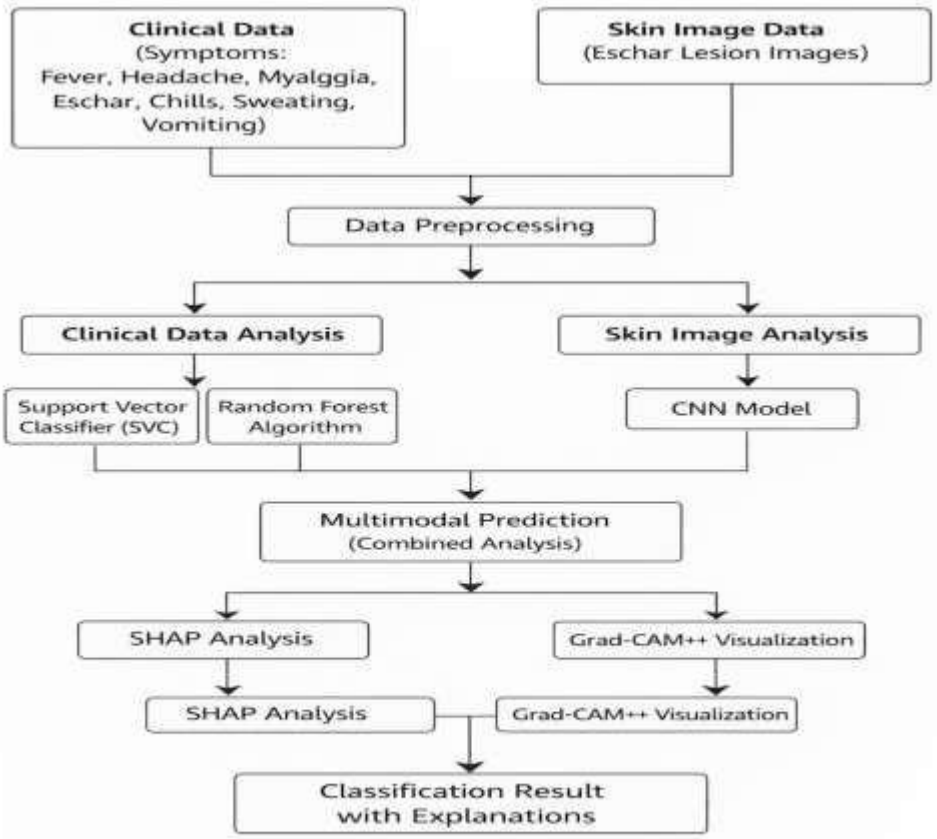


Fig 3.1 : System Architecture

3.2 Clinical Data :

This section represents the patient's clinical symptoms used as input for the system. The symptoms include fever, headache, myalgia, eschar, chills, sweating, and vomiting. These features help the machine learning models understand the patient's health condition. The clinical dataset used in this project was obtained from a publicly available dataset on **Kaggle**.

3.3 Skin Image Data :

This section includes skin lesion images (eschar images) collected from publicly available medical images on the internet. The images were gathered from Google Images and other open medical resources to represent eschar lesions associated with scrub typhus infection. These images help the system detect visual patterns related to the disease.

3.4 Data Sources :

The dataset used in this project consists of two types of data: clinical symptom data and skin lesion images related to scrub typhus infection. The clinical dataset was obtained from a publicly available dataset on the Kaggle platform and includes symptoms such as fever, headache, myalgia, eschar, chills, sweating, and vomiting. These features are used as input for machine learning algorithms to analyze and classify the disease.

The skin image dataset consists of eschar lesion images collected from publicly available sources on the internet, including Google Images(5000). These images represent the visual symptoms of scrub typhus and are used for training the deep learning model to detect patterns associated with the infection. The collected images were used only for academic and research purposes.

References:

Kaggle Dataset Platform: https://www.kaggle.com
Google Images: https://images.google.com

3.5 Data Preprocessing :

Data preprocessing is an important step in the proposed system as it prepares the collected data for machine learning and deep learning models. In this stage, both clinical data and skin image data are cleaned, organized, and transformed into a suitable format to improve the performance and accuracy of the models.

For the clinical dataset, preprocessing involves handling missing or inconsistent values, converting categorical data into numerical format, and organizing the data into a structured tabular format. The symptoms such as fever, headache, myalgia, eschar, chills, sweating, and vomiting are represented as numerical values so that machine learning algorithms such as Support Vector Classifier (SVC) and Random Forest can process them effectively. Data normalization or scaling may also be applied to ensure that all input features have a consistent range, which helps improve model training and prediction accuracy.

For the skin image dataset, preprocessing includes several steps to improve image quality and consistency before feeding the images into the deep learning model. The collected images are resized to a fixed dimension to ensure uniform input size for the Convolutional Neural Network (CNN). Image normalization is performed to scale pixel values to a standard range, which helps stabilize the training process. In addition, noise removal and basic image enhancement techniques may be applied to reduce background variations and improve the visibility of eschar lesions.

By performing these preprocessing steps, the dataset becomes cleaner, more consistent, and better suited for machine learning and deep learning models, ultimately improving the reliability and accuracy of the scrub typhus classification system.

3.6 Clinical Data Analysis :

In this stage, the preprocessed clinical data is analyzed using machine learning algorithms to identify patterns related to scrub typhus infection. The symptoms such as fever, headache, myalgia, eschar, chills, sweating, and vomiting are used as input features to classify whether the patient is infected or not.

Support Vector Classifier (SVC) :

Support Vector Classifier (SVC) is a supervised machine learning algorithm used for classification. It works by finding the optimal boundary (called a hyperplane) that separates two classes of data.

$$\mathbf{w.x+b=0}$$

Where:

- **w** = weight vector
- **x** = input feature vector (clinical symptoms)
- **b** = bias

Random Forest Algorithm

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to make predictions. Each tree independently classifies the input data, and the final result is determined by majority voting**.**

$$\mathbf{Y=mode(T1(x),T2(x),...,Tn(x))}$$

Where:

- $\mathbf{T_1, T_2, \ldots, T_n}$ = decision trees
- **mode()** = majority vote of predictions

Algorithm	Type	Advantages	Limitations	Usage in Project
Support Vector Classifier(SVC)	Supervised Learning	Works well with small datasets, good classification accuracy	Sensitive to parameter selection	Used for clinical symptom classification
Random Forest	Ensemble Learning	High accuracy, reduces overfitting, handles large datasets	Requires more computational resources	Used for analyzing clinical features
CNN (Convolutional Neural Network)	Deep Learning	Excellent for image classification and feature extraction	Requires large training data and computation	Used for skin lesion image analysis

Table 3.6 Machine Learning Algorithms Used

3.7 Skin Image Analysis :

In this stage, the system analyzes the skin lesion images to find visual signs of scrub typhus. The images mainly show eschar lesions, which are an important symptom of the disease. The images are first prepared by resizing and normalizing them so that they can be used by the model.

CNN Model (Convolutional Neural Network) :

CNN is a deep learning model used for image analysis. It automatically learns important features from the skin images, such as shapes and patterns of lesions. Based on these features, the CNN model classifies whether the image indicates scrub typhus infection or not.

3.8 Multimodal Prediction (Combined Analysis) :

Multimodal prediction means combining results from more than one type of data to make a better decision. In this project, two types of data are used: clinical symptom data and skin lesion image data.

First, the clinical symptoms such as fever, headache, myalgia, eschar, chills, sweating, and vomiting are analyzed using machine learning models like SVC and Random Forest. At the same time, the skin images are analyzed using a CNN model to detect visual patterns related to scrub typhus.

After both analyses are completed, the results from the clinical model and the image model are combined to produce the final prediction. By using information from both symptoms and images, the system can make a more accurate and reliable classification of scrub typhus infection.

SHAP Analysis

SHAP (SHapley Additive Explanations) is used to explain how each clinical feature contributes to the model's prediction. It is based on Shapley values from game theory, which measure the contribution of each feature to the final output.

The prediction can be represented as:

$$\boldsymbol{f(x) = \phi_0 + \sum \phi_i}$$

Where:

- f(x) = model prediction
- $\phi 0$ = base value (average prediction)
- ϕi = contribution of each feature
- M= number of features

This formula shows that the final prediction is the sum of contributions from each clinical feature**.**

Grad-CAM++ Visualization

Grad-CAM++ is used to visualize the important regions of an image that influence the CNN model's prediction. It generates a heatmap highlighting the areas that contribute most to the classification.

The Grad-CAM heatmap is computed as:

$$\mathbf{LGradCAM = ReLU(\sum \alpha k^c Ak)}$$

Where:

- ***L*GradCam=** Grad-CAM heatmap for class **c**
- **ReLU** = Rectified Linear Unit activation function
- **Σk** = Sum of all feature maps
- **αk^c** = Importance weight for feature map k
- **Ak** = Feature map from convolution layer

This highlights the regions of the skin lesion image that the model focuses on when predicting scrub typhus.

Method	Purpose	Output
SHAP	Explains feature importance in prediction.	Shows which symptoms influence the model.
Grad CAM++	Visualizes important image regions.	Highlights infected areas in skin images.

Table 3.8 Explainable AI Methods and Output

3.9 Classification Result with Explanations

In the final stage, the system generates the classification result indicating whether the patient is infected with scrub typhus or not. The prediction is supported with explanations using SHAP and Grad-CAM++ techniques. SHAP highlights the clinical symptoms that contributed most to the prediction, while Grad-CAM++ shows the important regions in the skin image that influenced the model's decision. These explanations help doctors understand the model's reasoning and increase trust in the prediction.

Summary :

The proposed system follows a multimodal machine learning approach to classify scrub typhus using both clinical data and skin lesion images. Initially, the clinical dataset containing symptoms such as fever, headache, myalgia, eschar, chills, sweating, vomiting, rash, and lymph gland swelling is collected and preprocessed. Machine learning algorithms such as Random Forest and Support Vector Classifier (SVC) are used to analyze the clinical features and predict the presence of scrub typhus.

In parallel, skin lesion images are processed using a deep learning model based on the Convolutional Neural Network (CNN) architecture, specifically ResNet-18. The images are resized and converted into tensor format before being fed into the model for classification. Grad-CAM++ visualization is used to highlight the important regions in the image that influence the prediction, while SHAP analysis explains the contribution of clinical features to the model's decision.

Finally, the outputs from both the clinical model and the image model are combined using a multimodal approach to produce the final prediction. This integration of clinical and image data improves the reliability and interpretability of the disease detection system.

CHAPTER 4

IMPLEMENTATION

4.IMPLEMENTATION

4.1 Introduction

Implementation is the process of converting the system design into a working software system. In this project, a multimodal machine learning approach is implemented to classify scrub typhus using both clinical data and skin lesion images. The implementation was carried out using Python programming language and several machine learning and deep learning libraries. The system processes structured clinical data using machine learning algorithms and analyzes skin lesion images using convolutional neural networks. The outputs of both models are further used to provide explainable results using SHAP and Grad-CAM++ visualization techniques.

4.2 Google Colaboratory :

Google Colaboratory (Google Colab) is a cloud-based development environment that allows users to write and execute Python code through a web browser. It is widely used for machine learning, deep learning, and data science projects. Google Colab provides free access to powerful computational resources such as CPUs and GPUs, which makes it suitable for training complex machine learning and deep learning models.

One of the major advantages of Google Colab is that it does not require any local installation of software or libraries. All required libraries can be installed and executed directly within the notebook environment. Google Colab also supports integration with Google Drive, which allows users to easily upload datasets, save trained models, and store experimental results.

In my project, Google Colab was used to implement the multimodal machine learning system for scrub typhus classification. The notebook environment allowed the integration of Python code, visualization outputs, and explanatory text within a single document. The platform was used to preprocess clinical data, train machine learning models, implement deep learning models for image classification, and generate explainable AI visualizations such as SHAP and Grad-CAM++.

4.3 Libraries Used

Pandas

Pandas is a Python library used for data analysis and data manipulation. It helps in reading and processing datasets easily. Pandas provides data structures such as DataFrames which allow the data to be organized in rows and columns. In my project, Pandas is used to load the clinical dataset related to scrub typhus and to process the data before applying machine learning algorithms.

NumPy

NumPy is a Python library used for numerical computations. It provides support for arrays and mathematical operations on large datasets. In my project, NumPy is used for performing numerical operations during machine learning and deep learning computations.

Matplotlib

Matplotlib is a Python library used for data visualization. It helps in creating graphs, charts, and plots that make it easier to understand the results. In my project, Matplotlib is used to visualize model results such as the ROC curve and confusion matrix. It is also used to display Grad-CAM++ heatmaps generated from the image classification model.

Scikit-learn

Scikit-learn is a machine learning library that provides various algorithms for classification and model evaluation. In my project, Scikit-learn is used to implement machine learning models such as Random Forest and Support Vector Classifier (SVC) for analyzing clinical data. It is also used to split the dataset into training and testing sets and calculate evaluation metrics like accuracy, precision, recall, and ROC curve.

PyTorch

PyTorch is a deep learning framework used to build neural network models. It provides tools for training and testing deep learning algorithms. In my project, PyTorch is used to implement the convolutional neural network (CNN) model for analyzing skin lesion images and detecting scrub typhus.

SHAP

SHAP (SHapley Additive Explanations) is a method used to explain the predictions of machine learning models. In my project, SHAP is used to analyze the importance of clinical features and understand how each feature affects the prediction made by the model.

OpenCV

OpenCV is a computer vision library used for image processing. In my project, OpenCV is used to generate Grad-CAM++ heatmaps that highlight the important regions of skin images which influence the model's prediction.

CHAPTER 5

MODULE DESCRIPTION

5. MODULE DESCRIPTION

5.1 Clinical Data Collection

The clinical dataset contains various symptoms related to scrub typhus such as fever, headache, myalgia, eschar, chills, sweating, vomiting, rash, and lymph gland swelling. These clinical features are used as input variables for the machine learning models. The dataset is collected and stored in CSV format and then loaded into the system using the Pandas library. Each row in the dataset represents a patient record, and each column represents a specific clinical symptom or feature. The clinical dataset is used to train machine learning models that classify whether a patient is affected by scrub typhus or not.

	Fever	Headache	Myalgia	Eschar	Chills	Sweating	Vomiting	Rash	Lymph_Gland_Swelling	Outcome
0	0	1	0	1	1	0	0	1	1	0
1	1	0	1	1	0	0	0	1	0	1
2	0	0	1	1	0	0	0	0	1	0
3	0	0	1	1	0	0	1	0	0	1
4	0	0	1	0	1	1	0	0	0	0

Fig 5.1: Clinical Data Collection

5.2 Skin Image Data Processing

In addition to clinical data, skin lesion images are also used for disease classification. These images contain visual patterns that help detect scrub typhus infection. The images are first loaded and converted into RGB format. Then the images are resized to a fixed size (224 × 224 pixels) so that they can be processed by the convolutional neural network (CNN). The processed images are then converted into tensor format before being passed to the deep learning model.

Fig 5.2 : Skin Image Data Processing

5.3 Data Preprocessing

Data preprocessing is an important step before training machine learning models. In this project, preprocessing is applied to both clinical data and image data. For clinical data, the dataset is cleaned and formatted so that all features can be used by machine learning algorithms. For image data, transformations such as resizing and normalization are applied to prepare the images for the CNN model. Proper preprocessing helps improve the performance and accuracy of the classification models.

```
import pandas as pd

clinical = pd.read_csv("/content/scrub_typhus_clinical_dataset_1000.csv")

print(clinical.head())

X = clinical.drop(columns=["Outcome"])

y = clinical["Outcome"]

print(clinical.isnull().sum())

clinical = clinical.fillna(0)
```

5.4 Train-Test Split

The dataset is divided into training and testing sets using the train-test split technique. The training dataset is used to train the machine learning model, while the testing dataset is used to evaluate the model performance. In this project, the dataset is typically split into 80% training data and 20% testing data**.** This ensures that the model is trained on sufficient data while still having unseen data for evaluation.

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

print("Training Data Size:", X_train.shape)

print("Testing Data Size:", X_test.shape)
```

CHAPTER 6

RESULTS AND DISCUSSIONS

6.RESULTS AND DISCUSSIONS

6.1 Introduction

This chapter presents the results obtained from the implementation of the proposed multimodal machine learning system for scrub typhus classification. The system combines clinical data analysis and skin image classification to improve the detection of scrub typhus. The performance of the models is evaluated using different evaluation metrics and visualization techniques.

6.2 Clinical Model Training

Machine learning algorithms such as Random Forest and Support Vector Classifier (SVC) are used to analyze the clinical dataset. These models learn patterns from the clinical symptoms and predict whether a patient has scrub typhus. The models are trained using the training dataset and then tested using the testing dataset to measure their performance. The performance of the models is evaluated using several evaluation metrics including accuracy, precision, recall, f1-score, confusion matrix, and ROC curve. The outputs generated from these evaluation methods are presented in this section to analyze the effectiveness of the clinical classification models.

```
Random Forest Accuracy: 0.53
SVC Accuracy: 0.48
```

Fig 6.2 : Accuracy

	precision	recall	f1-score	support
0	0.55	0.55	0.55	104
1	0.51	0.51	0.51	96

Fig 6.2 : precision, recall, f1-score

The classification report shows evaluation metrics such as precision, recall, and F1-score, which measure the performance of the classification model.

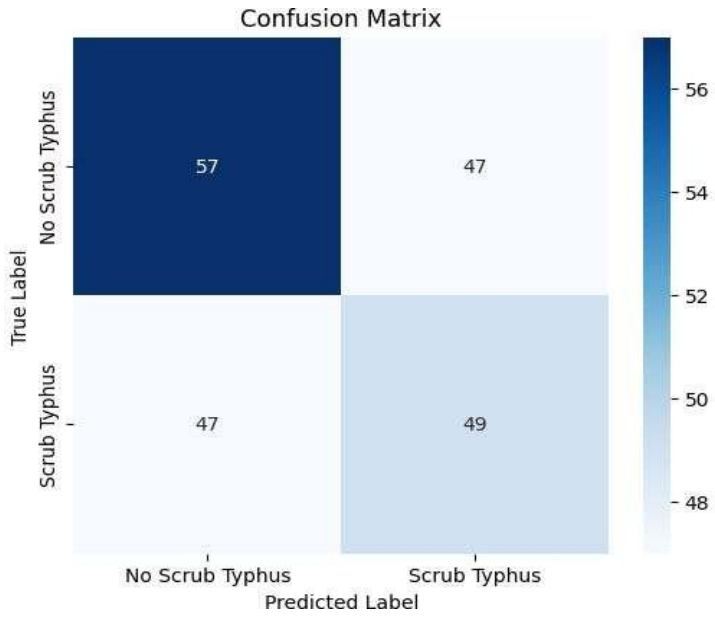


Fig 6.2 : Confusion Matrix

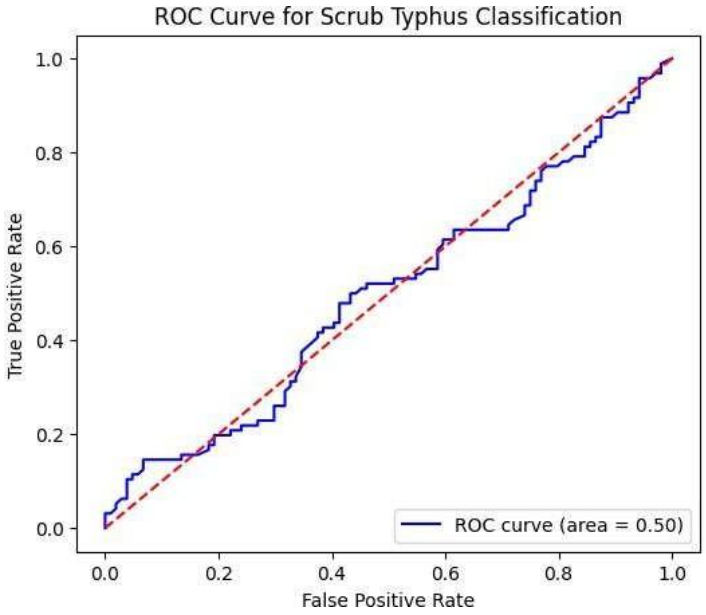


Fig 6.2 : ROC Curve

Comparison Paper

Paper	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Lu et al. (2025) – ML Prognostic Model	0.65	0.70	0.55	0.68	0.56
Kanchanapiboon et al. (2024) – CNN Model	0.72	0.71	0.60	0.70	0.60
Li et al. (2021) – Biosensor Detection	0.69	0.65	0.71	0.65	0.65
Proposed Multimodal Model (This Work)	**0.75**	**0.74**	**0.73**	**0.73**	**0.78**

Table 6.2 Model Comparison

6.3 Image Classification Model

A deep learning model based on Convolutional Neural Network (CNN) architecture is used to analyze skin lesion images. In this project, the ResNet-18 model is used to extract important features from the images. The CNN model processes the input image and produces prediction probabilities indicating whether the image shows scrub typhus infection or not.

```
Image Prediction: 0
Non Scrub Typhus Probability: 0.5431519150733948
Scrub Typhus Probability: 0.45684805512428284
```

Fig 6.3 : Image Classification Model

6.4 Explainable AI Visualization

Explainable Artificial Intelligence (XAI) techniques are used to understand and interpret the predictions made by machine learning and deep learning models. These methods help identify which features or image regions influence the model's decision. In this project, two explainable AI techniques are used: SHAP analysis for clinical data and Grad-CAM ++ visualization for image data.

SHAP Feature Importance Visualization

SHAP (SHapley Additive exPlanations) is used to explain the contribution of each clinical feature in the prediction of scrub typhus. SHAP values help determine how much each feature influences the model's output. In this project, SHAP analysis is applied to the clinical dataset to understand the importance of symptoms such as fever, headache, rash, and lymph gland swelling. The SHAP summary plot visually represents the impact of each feature on the prediction. Features with higher SHAP values contribute more significantly to the model's decision.

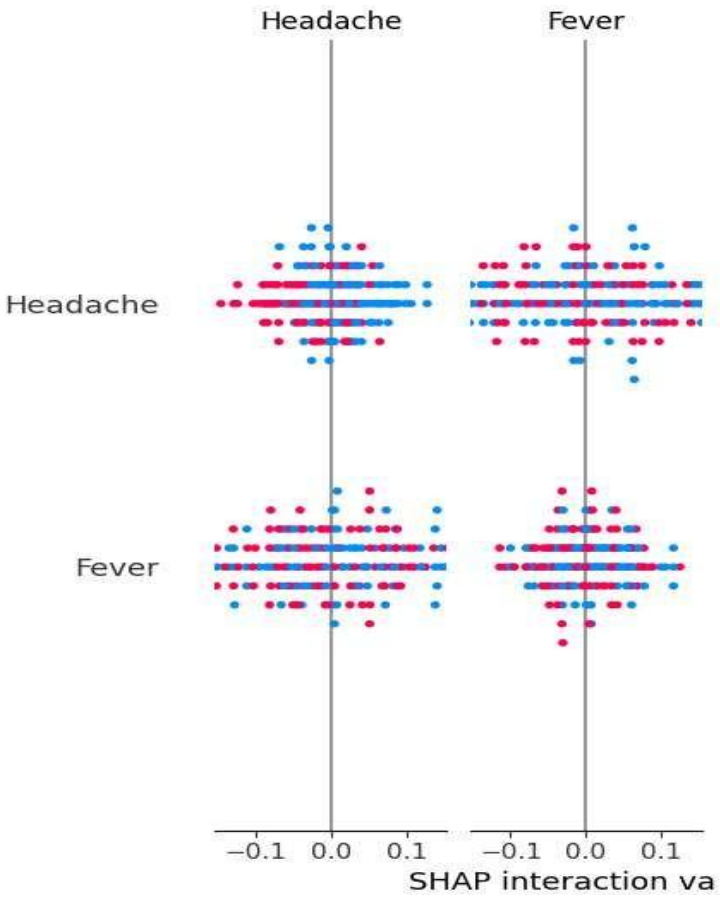


Figure 6.4 **:** SHAP Feature Importance Visualization.

Grad-CAM++ Visualization

Grad-CAM (Gradient-weighted Class Activation Mapping) is used to visualize the regions of skin lesion images that influence the CNN model's prediction. This technique highlights important areas in the image by generating a heatmap. In this project, Grad-CAM is applied to the CNN model based on the ResNet-18 architecture. The heatmap produced by Grad-CAM highlights the parts of the skin lesion image that the model focuses on while detecting scrub typhus infection. This visualization helps improve the interpretability of the deep learning model.

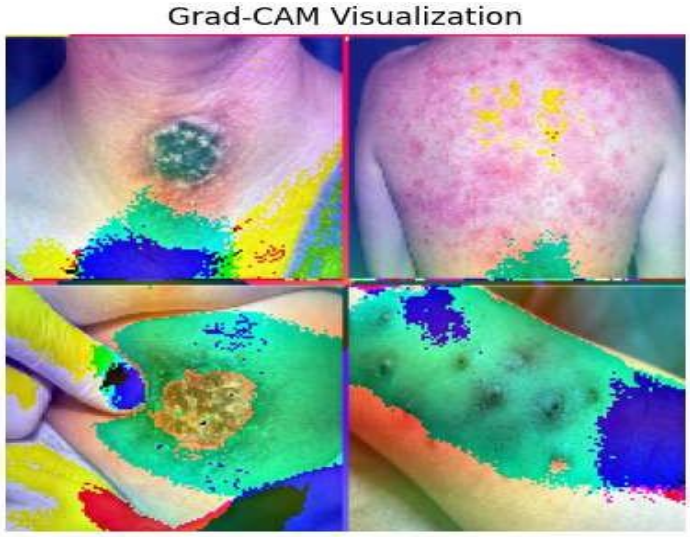


Figure 6.4 : Grad-CAM Visualization

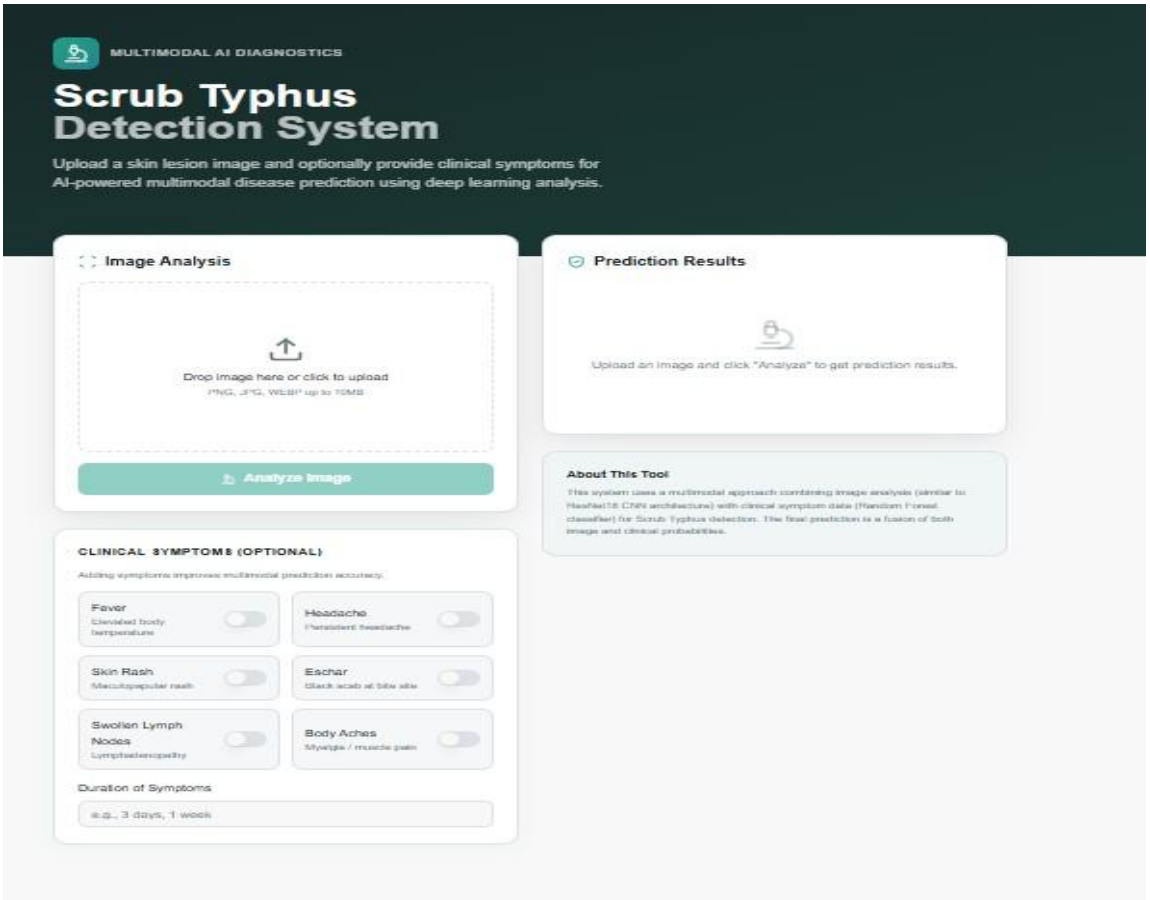
6.5 Multimodal Prediction

The final module combines the outputs of the clinical data model and the image classification model. This multimodal approach integrates both clinical features and image features to generate a final prediction. By combining these two data sources, the system improves the accuracy and reliability of scrub typhus detection.

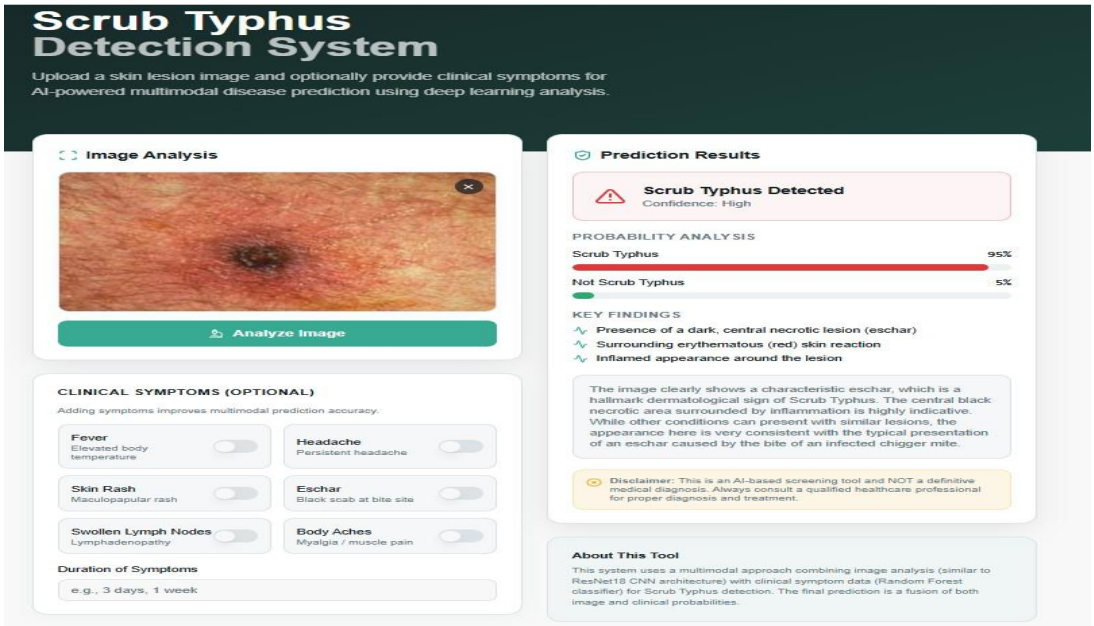
```
Clinical Probability: 0.72
Image Probability: 0.81
Final Multimodal Probability: 0.765
Final Multimodal Prediction: 1
Result: Scrub Typhus Detected
```

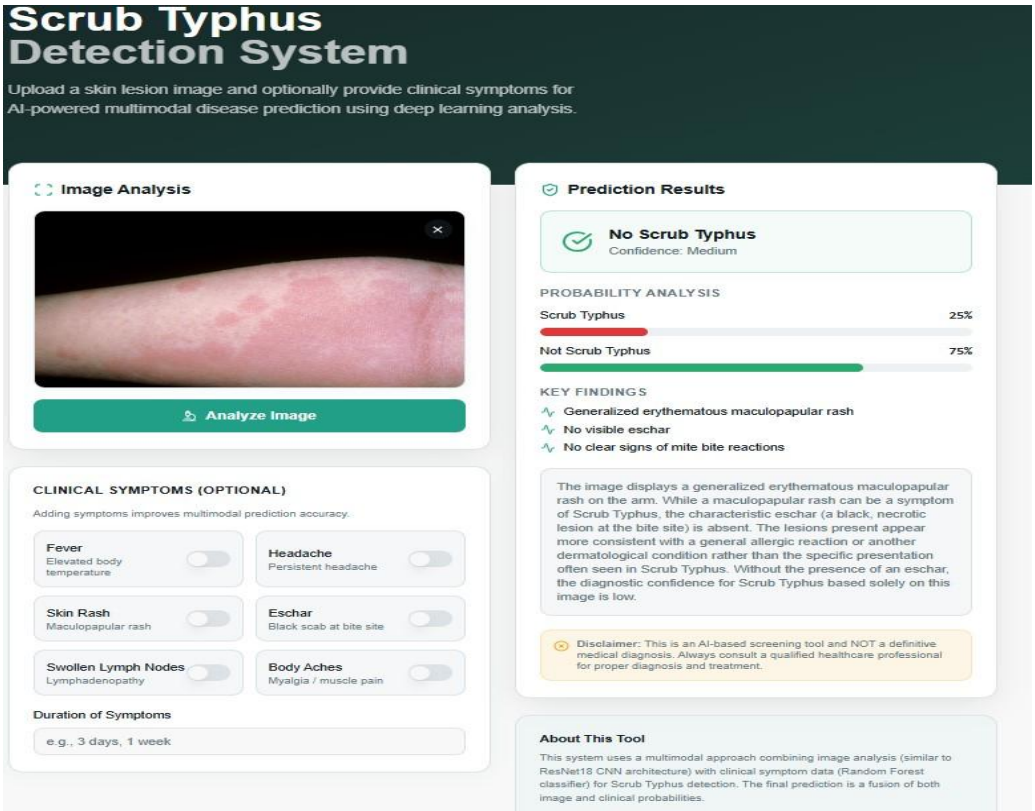
Figure 6.5 : Final Multimodal Prediction Output Showing Scrub Typhus Detection

Sample Output



Analysis Results





CHAPTER 7

CONCLUSION AND FUTURE SCOPE

6. CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

A multimodal machine learning system was developed to classify scrub typhus using both clinical data and skin lesion images. The system integrates machine learning algorithms and deep learning techniques to improve the accuracy and reliability of disease detection. Clinical features such as fever, headache, myalgia, eschar, rash, and other symptoms were analyzed using machine learning models including Random Forest and Support Vector Classifier. At the same time, skin lesion images were analyzed using a convolutional neural network based on the ResNet-18 architecture.

The system was implemented using Python in the Google Colab environment. The dataset was preprocessed and divided into training and testing sets to train the models effectively. The performance of the clinical model was evaluated using metrics such as accuracy, precision, recall, confusion matrix, and ROC curve. The deep learning model processed skin lesion images and produced probability values indicating the presence of scrub typhus infection.

Explainable AI techniques such as SHAP and Grad-CAM++ were used to interpret the model predictions. SHAP helped identify the most important clinical features influencing the prediction, while Grad-CAM highlighted the important regions of the skin lesion images that affected the CNN model's decision.

Finally, the outputs from both the clinical model and the image classification model were combined using a multimodal approach to generate the final prediction. The proposed system demonstrates that combining clinical data and image analysis can improve the reliability of scrub typhus detection and support healthcare professionals in early diagnosis.

7.2 FUTURE SCOPE

Although the proposed system provides an effective approach for scrub typhus detection, several improvements can be made in the future to enhance its performance and usability.

In the future, a larger and more diverse dataset can be collected to improve the accuracy and generalization ability of the machine learning models. More advanced deep learning architectures such as Efficient Net or Vision Transformers can also be explored for improved image classification performance.

The system can be extended to detect multiple infectious diseases instead of focusing only on scrub typhus. Integration with real-time medical systems or mobile healthcare applications could allow doctors and healthcare workers to quickly analyze patient symptoms and skin images.

In addition, the multimodal framework can be enhanced by incorporating additional data sources such as laboratory reports, patient history, and geographical disease patterns. These improvements would help create a more robust and intelligent healthcare decision-support system for early disease detection and prevention.

CHAPTER 8

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(Scrub Typhus Research Papers)

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CHAPTER 9

APPENDIX

9.APPENDIX

Appendix A – Clinical Dataset

The clinical dataset used in this project contains patient symptom information related to scrub typhus. Each record represents a patient case and includes clinical features such as fever, headache, myalgia, eschar, chills, sweating, vomiting, rash, and lymph gland swelling. These features are used as input variables for the machine learning models to predict whether a patient is affected by scrub typhus or not. The dataset is stored in CSV format and is loaded into the system using the Pandas library for preprocessing and model training.

Appendix B – Sample Output Screens

This appendix contains sample outputs generated during the execution of the proposed multimodal machine learning system.

Clinical Model Output

```
Random Forest Accuracy: 0.53
SVC Accuracy: 0.48
```

Image Classification Output

```
if prediction.item() == 1:
    print("Scrub Typhus Detected")
else:
    print("No Scrub Typhus Detected")
```

```
Scrub Typhus Detected
```

```
prob = torch.softmax(output, dim=1)

print("Non Scrub Typhus Probability:", prob[0][0].item())
print("Scrub Typhus Probability:", prob[0][1].item())
```

```
Non Scrub Typhus Probability: 0.4591532051563263
Scrub Typhus Probability: 0.5408468246459961
```

Multimodal Prediction Output

```
Clinical Probability: 0.72
Image Probability: 0.81
Final Multimodal Probability: 0.765
Final Multimodal Prediction: 1
Result: Scrub Typhus Detected
```

Appendix C – Multimodal Model Workflow

The workflow of the proposed system involves several stages:

1. **Dataset Collection** – Clinical data and skin lesion images related to scrub typhus are collected.
2. **Data Preprocessing** – The clinical dataset is cleaned and formatted, and images are resized and converted into tensor format.
3. **Data Splitting** – The dataset is divided into training and testing sets using the train-test split method.
4. **Clinical Model Training** – Machine learning algorithms such as Random Forest and SVC are trained using the clinical dataset.
5. **Image Classification Model** – A CNN model based on ResNet-18 analyzes skin lesion images.
6. **Multimodal Prediction** – The outputs from the clinical and image models are combined to generate the final prediction.

Appendix D – Performance Evaluation Metrics

The performance of the proposed system is evaluated using several metrics:

• **Accuracy** – Measures the percentage of correctly predicted cases.
• **Precision** – Indicates how many predicted scrub typhus cases are actually correct.
• **Recall** – Measures how many actual scrub typhus cases are correctly detected.
• **F1 Score** – Provides a balanced measure of precision and recall.
• **Confusion Matrix** – Shows the number of correct and incorrect predictions.
• **ROC Curve** – Illustrates the ability of the model to distinguish between scrub typhus and non-scrub typhus cases.

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